Focused History
Obtain gestation, parity, prenatal care and previous complications if any

Probable Labor
Consider requesting a maternity ambulance.
Requesting a maternity ambulance is optional. Any clinical staff member regardless of gender may assist with delivery with patient consent.

Non-Imminent Birth

- Apply absorbent pad to perineum area
- Record timing of uterine contraction
- Monitor continuously

Imminent Birth
Consider staying on scene to deliver infant

Consider ALS Back-Up

Position Mother
Support mothers' preference (semi-recumbent, "all-fours", lateral)

Complicated Presentation

- Nuchal Cord
- Should Dystocia
- Prolapsed Cord
- Breech Birth
- Compound Presentation
- Cord Rupture
- Shoulder Dystocia

Uncomplicated Presentation
Cephalic Presentation

- Gently guide anterior shoulder under symphysis pubis, then posterior
- Check for nuchal cord (cord around neck); slip over head

Controlled and guided delivery of infant

Dry, warm and **stimulate infant**
1st APGAR assessment

Infant Stable

Delay cord cutting (>1min post-delivery)

2nd APGAR assessment

Assess mother
Consider post partum hemorrhage

Infant Unstable

Go to Neonatal Resus CPG

Nuchal Cord
(Cord Around Neck)

- After the head has been delivered, palpate the neck for a nuchal cord. If present, slip over the head.
- If the loop is too tight to slip over the head, attempt to slip the cord over the shoulders and deliver the body through the loop

Cord Rupture

- Apply direct pressure to the tear with a sterile dressing.
- Position the clamp proximally to tear, if possible, to control bleeding.

Cord Prolapse

Palpable and visible umbilical cord with ruptured membranes

Birth Imminent

Prepare for childbirth and/or newborn resuscitation

Birth Not Imminent

Position Mother
All-fours position
OR
Exaggerated SIMS position

- Insert fingers into vagina and gently push the presenting part (head) away from the cord
- Minimize handling of the cord
- Maintain pressure until birth commences or advised to release at hospital

If cord prolapsed outside vagina, Cover with pad soaked in warm saline

AP

Consider Tocolysis:
Nebulized Salbutamol
10-20mg Nebulized

Shoulder Dystocia

Suspect shoulder dystocia when:

- Difficulty with birth of the face and chin
- The fetal head retracts against the perineum (Turtle neck sign)
- Failure of the fetal head to restitute
- Failure of the shoulders to descend
- If fetal shoulders fail to deliver within 60 seconds post fetal head.

Once shoulder dystocia is identified, discourage pushing from the mother

Transport must be prioritized wherever possible with maneuvers (below) continued en route.

Position Mother
Place mother supine with her knees to chest position (McRoberts maneuver)

McRoberts with Suprapubic Pressure
If McRoberts maneuver fails:

- Keep mother in the same position and
- Use the heel of your hand to push on the mothers pubic symphysis, directing pressure downward and laterally to dislodge the anterior shoulder.

If the above fails,
Attempt to deliver the posterior arm:
Insert hand into the vagina, locate the posterior arm and sweep it across the chest to deliver it, which may rotate the shoulders.

If the above fails, perform Internal rotation maneuvers:
Rubins II:

- Insert fingers of one hand vaginally behind the posterior aspect of the anterior shoulder of the fetus and rotating the shoulder toward the fetal chest. OR/

Woods corkscrew:

- Place at least 2 fingers on the anterior aspect of the fetal posterior shoulder, applying gentle upward pressure around the circumference of the arc in the same direction as with the Rubin II maneuver above.

If above maneuvers fail,
Consider positioning mother in all fours position
Repeat all the above maneuvers

Breech Birth

Transport must be prioritized wherever possible with maneuvers (below) continued en route.

Position Mother

- Place mother supine, allow the buttocks, feet, and trunk to deliver spontaneously, then support the body while the head is delivered.
- If needed, put the mother in a kneeling position which may assist in the delivery of the newborn.

If the head fails to deliver

- Place gloved hand into vagina with fingers in between infant's face and uterine wall to create an open airway.
- Place index and ring fingers on the baby's cheeks forming a "V" taking care not to block the mouth and allowing the chin to tilt toward the chest flexing the neck.

- When delivering breech, you may need to rotate the baby's trunk clockwise or sweep the legs from the vagina.
- Once the legs are delivered support the body to avoid hyperextension of the head; keep the fetus elevated off the umbilical cord.
- NEVER pull on the body, especially a preterm or previable baby – just support the baby's body while mother pushes when she feels the urge to

Compound Presentation
(presentation of an arm or leg)

Prioritize Transport

Excessive Bleeding During Active Labor

- Place large bore IV and administer IV fluids as per PV Hemorrhage in Pregnancy CPG.

- **C-Section most likely needed – expedite transport**

Patient Care Goals

1. Obtain the necessary history to plan for the birth and resuscitation of the newborn.
2. Recognize imminent birth.
3. Assist with the uncomplicated delivery of term newborns.
4. Recognize complicated delivery situations (e.g., nuchal or prolapsed umbilical cord, breech delivery, shoulder dystocia)
5. Apply appropriate techniques when an obstetric complication exists.
6. Safely manage imminent and non-imminent childbirth to ensure the health and safety of both mother and infant.
7. Facilitate delivery with minimal complications and provide timely resuscitation if required.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients with signs of imminent birth with crowning:
  - Painful contractions lasting 45-60 seconds at 1–2-minute intervals:
  - Bearing down/urge to open bowels.
  - Bloody show, bulging perineum, fetal head crowning, or membrane rupture.
- Patients with non-imminent birth:
  - Irregular contractions, timing, and strength.
  - Abdominal tightening without pain
  - No bloody show or visible infant body parts at the vulva.
  - Membranes intact.

Exclusion Criteria

- Vaginal bleeding in any stage of pregnancy (See PV Hemorrhage in Pregnancy CPG)
- Emergencies in the first or second trimester of pregnancy.

Key Patient Assessment Points

1. **Identify Features of Active Labor/Imminent Delivery**
   - Increasing frequency and severity of contractions with an urge to push or bear down.
   - Membrane rupture (note color of amniotic fluid if ruptured).
   - Loss of mucous plug or bloody show.
   - Bulging perineum or appearance of presenting part at the vulva.
2. **Identify Delivery Complications**
   - Cord presentation or prolapse.
   - Breech birth.
   - Shoulder dystocia.
   - Nuchal Cord.
3. **Determine Obstetric History**
   - Onset of labor symptoms (e.g., timing, frequency, and duration of contractions).
   - Gestational age.
   - Gravida/para status (number of pregnancies and births).
   - Potential complications or risk factors (e.g., gestational diabetes, pre-eclampsia, multiple pregnancy, mal-presentation, previous childbirth, obstetric or gynecological problems (e.g., shoulder dystocia, postpartum hemorrhage).
   - Concurrent symptoms (e.g., pain level, vaginal bleeding, fever).
   - Treatments or actions prior to EMS arrival (e.g., analgesia, positioning, any home interventions).
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority:
       - Imminent Delivery (until baby delivered without complication)
       - Complications in pregnancy (e.g. prolapsed cord, presenting limb).

Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for:
     - Imminent Delivery:
       - Note that ALS-Back-up can be requested and cancelled if not required.
       - If ALS back-up is not available a second ambulance (of EMT or Paramedic level) should be routinely requested.
     - Delivery Complications.
1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
   - Determine obstetric history.
2. **If No Signs if Imminent Delivery**
   - If the patient is in labor but has no signs of imminent delivery, transport to the appropriate receiving facility.
3. **Active Labor / Imminent Delivery**
   - Delivery should be controlled to allow a slow, controlled delivery of the infant. This will prevent injury to the mother.
   - Do not routinely suction the infant's airway (even with bulb syringe) during delivery.
   - **Deliver the Newborn:**
     - Support infant's head as needed and apply gentle counterpressure to help prevent the head from suddenly popping out.
     - Grasping the head with hand over the ears, gently guide head down to allow delivery of the anterior shoulder.
     - Gently guide the head up to allow delivery of the posterior shoulder.
     - Slowly deliver the remainder of the infant.
     - After 1 minute, clamp the cord in three places (10cm, 15cm, 20cm) from the abdomen; cut the cord between the last two clamps.
       - Whild cord is attached, take care to ensure baby is not significantly higher positioned than the mother to prevent blood from flowing backwards from baby to placenta.
       - **If resuscitation is needed:**
         - The baby can still benefit from a 1-minute delay in cord clamping. Start resuscitation immediately after birth and then clamp and cut the cord at 1-minute.
     - Dry, warm, and stimulate infant, wrap in towel and place on maternal chest unless resuscitation is needed.
   - **Record APGAR Score:**
     - Record APGAR scores at 1 and 5 minutes once the neonate is stabilized.
     - Resuscitation takes priority over recording APGAR scores.

	Sign	2	1	0
A	**Appearance** (skin color)	Normal over entire body	Normal except extremities	Cyanotic or pale all over
P	**Pulse**	>100 bpm	<100 bpm	Absent
G	**Grimace** (reflex irritability)	Sneezes, coughs or vigorous cry	Grimaces	No response
A	**Activity** (Muscle tone)	Active	Arms and legs flexed	Absent
R	**Respirations**	Good, crying	Gasping, irregular	Absent

4. **After Delivery - Mother**

- After delivery of infant, routine suctioning (including bulb suctioning) should be reserved for infants who have obvious obstruction to the airway or require positive pressure ventilation (Refer to Neonatal Resuscitation CPG).
- **Placental Delivery:**
  - The placenta will deliver spontaneously, often within 5-15 minutes after the infant is delivered.
    - Do not force the placenta to deliver, do not pull on the umbilical cord
    - Place placenta in biohazard bag and transport with the mother.
- **Monitor for Post Partum Hemorrhage**
  - Estimate maternal blood loss
  - **Uterine Massage:**
    - After delivery, massaging the uterus (located at about the umbilicus) and allowing the infant to breastfeed will promote uterine contraction and help control bleeding.
    - Treat mother for hypovolemia as needed (See Post-Partum Hemorrhage CPG).

5. **After Delivery - Child**
   - **Cord Clamping**
     - Cutting the cord should be delayed by 1-3 minutes.
     - Clamp cord in 3 places: (10cm, 15cm, 20cm from umbilicus)
       - Cut between 2 clamps furthest away from the newborn
     - Cord milking should be avoided.
   - Keep the infant warm during transport.
     - Kangaroo care/skin-to-skin is suitable.
     - Place towel or blanket over mother and infant where possible.
6. **Delivery Complications**
   - Most deliveries proceed without complications – if complications of delivery occur:
     - Apply high flow oxygen (15Lpm via NRB mask) to mother and expedite transport. Maternal resuscitation is critical for best fetal outcome.
   - **Shoulder Dystocia**
     - If delivery fails to progress after head delivers, attempt the following;
     - McRoberts' maneuver:
       - Hyperflex the mother's hips to severe supine knee-chest position
       - Apply firm suprapubic pressure to attempt to dislodge the shoulder. This often requires two EMS clinicians to perform and allows for delivery in up to 70% of cases
       - Attempt to angle baby's head as posteriorly as possible but NEVER pull
       - Continue with delivery as normal once the anterior shoulder is delivered.
   - **Nuchal Cord** (cord wrapped around the baby's neck)
     - After the head has been delivered, palpate the neck for a nuchal cord. If present, slip over the head.
     - If the loop is too tight to slip over the head, attempt to slip the cord over the shoulders and deliver the body through the loop
   - **Cord Rupture**
     - Apply direct pressure to the tear with a sterile dressing.
     - Position the clamp proximally to tear, if possible, to control bleeding.
   - **Prolapsed Cord**
     - Placed gloved hand into vagina and gently lift head/body off the cord
     - Assess for pulsations in cord, if no pulses are felt, lift the presenting part off the cord.
     - Wrap the prolapsed cord in moist sterile gauze
     - Maintain until relieved by hospital staff
     - If previous techniques are not successful, place mother in the prone knee-chest position or extreme Trendelenburg with hips elevated.
     - **Consider Tocolysis:**
       - Criteria:
         - Prolapsed Cord, and/

        - Birth is not imminent.
    - **Administer Nebulized Salbutamol:**
        - Nebulized Salbutamol: 10mg-20mg.
- **Shoulder Dystocia**
    - General Approach:
        - Once shoulder dystocia is identified, discourage pushing from the mother.
        - Transport must be prioritized wherever possible, with maneuvers continued en route if needed.
        - **Supportive Interventions** (Follow HELPR Mnemonic)
            - H (Help): Call for help (ALS Back-Up).
            - E (Legs): **Position mother in McRoberts' maneuver** (hyperflex mother's hips to severe supine knee-chest position).
            - L (Pressure): **Apply firm suprapubic pressure to dislodge the shoulder** (often requires two EMS clinicians and allows delivery in up to 75% of cases).
            - E (Enter Maneuvers): If needed, perform internal rotational maneuvers or posterior arm delivery; consult Tele EMS if consulting rotational maneuvers. See below.
            - P (Remove Posterior Arm): Attempt to angle baby's head as posteriorly as possible but NEVER pull; if the above fails, insert hand into the vagina, locate the posterior arm, and sweep it across the chest to deliver it, which may rotate the shoulder.
            - R (Roll): Roll the patient onto all fours if initial maneuvers fail.
        - **Internal Rotational Maneuvers**
            - Consult Tele-EMS prior to attempting Internal Rotational Maneuvers.
            - Rubins II:
                - Insert fingers of one hand vaginally behind the posterior aspect of the anterior shoulder of the fetus and rotating the shoulder toward the fetal chest. OR/
            - Woods corkscrew:
                - Place at least 2 fingers on the anterior aspect of the fetal posterior shoulder, applying gentle upward pressure around the circumference of the arc in the same direction as with the Rubin II maneuver above.
- **Breech Birth**
    - Place mother supine, allow the buttocks, feet, and trunk to deliver spontaneously, then support the body while the head is delivered.
    - If needed, put the mother in a kneeling position which may assist in the delivery of the newborn.
    - Assess for presence of prolapsed cord and treat as above.
    - If the head fails to deliver, place gloved hand into vagina with fingers in between infant's face and uterine wall to create an open airway. Place index and ring fingers on the baby's cheeks forming a "V" taking care not to block the mouth and allowing the chin to tilt toward the chest flexing the neck.
    - When delivering breech, you may need to rotate the baby's trunk clockwise or sweep the legs from the vagina.
    - Once the legs are delivered support the body to avoid hyperextension of the head; keep the fetus elevated off the umbilical cord.
    - NEVER pull on the body, especially a preterm or previable baby – just support the baby's body while mother pushes when she feels the urge to
- **Compound Presentation** (presentation of an arm or leg)
    - The presentation of an arm or leg through the vagina is an indication for immediate transport to the hospital.
- **Excessive Bleeding During Active Labor**
    - Excessive bleeding during active labor may occur with placenta previa or placental abruption.
        - Obtain obstetric history from patient.
        - Placenta previa most likely will prevent delivery of the infant vaginally
        - Place large bore IV and administer IV fluids as per PV Hemorrhage in Pregnancy CPG.
        - C-Section most likely needed – expedite transport

7. **Maternal Cardiac Arrest**
   - Apply manual pressure to displace uterus from midline (to the left).
   - Treat per Adult Cardiac Arrest CPG.
   - If infant is estimated to be over 24 weeks gestation, rapid transfer to hospital is required for Perimortem C-Section (Peri-mortem C-Section is most successful if started within 5 minutes of maternal cardiac arrest).
8. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Mother:
       - Vital signs every 5–10 minutes.
       - Monitor for signs of hypovolemia or uterine atony.
     - Infant:
       - APGAR Score at 1 and 5 minutes.
       - Warmth, respiratory effort, tone, APGAR monitoring.
       - Observe for need for resuscitation or oxygen escalation

Patient Safety Considerations

- Supine Hypotension Syndrome:
  - If mother has hypotension before delivery, place patient in left lateral recumbent position or manual displace gravid uterus to the left in supine position
- Knee-chest position may create safety issues during rapid ambulance transport.
- Do not routinely suction the infant's airway during delivery (even with bulb syringe).
- Newborns are very slippery, take care not to drop the infant.
- Dry, warm and stimulate newborns to facilitate respirations and prevent hypothermia.
- Do not pull on the umbilical cord while the placenta is delivering.
- If possible, transport between deliveries if the mother is expecting twins.

Clinical Insights & Additional Information

- Determining Between Imminent Birth and Non-Imminent Birth
  - Imminent Birth
    - Painful contractions: 45-60 secs at 1-2 min intervals
    - Bearing down/urge to open bowels
    - Bloody show
    - Bulging perineum, fetal head crowning, membrane rupture
  - Non-Imminent Birth
    - Irregular contractions, timing and strength
    - Abdominal tightening without pain
    - No bloody show
    - No infant body parts visible at vulva
    - Membranes intact
- Requesting Back-up:
  - Back-up should be routinely requested for imminent birth.
  - Ideally back-up should be ALS (paramedic or above) however, if no ALS is easily available, a second BLS (EMT) unit should be requested incase neonatal resuscitation is required, or the mother requires treatment.
- It is acceptable for a male clinician to assist in delivery of a newborn – providing this is acceptable to the patient.
  - If the patient does not consent, it is appropriate to request a maternity ambulance (MAMA unit).
  - Routine calling for a maternity ambulance should be avoided.

- Dispatch 2 co-ordination:
  - Ensure Dispatch 2 is notified of the following conditions when determining receiving hospital:
    - Antepartum hemorrhage
    - Post partum hemorrhage
    - Breech presentation
    - Limp presentation
    - Complicated nuchal cord
    - Prolapsed cord
    - Shoulder dystocia
    - Maternal cardiac arrest

- All Fours Position:

  - Top: “all fours” position.
  - Bottom: Exaggerated SIMS position

- DCAS Delivery Kit (Obstetric Kit) Contents:

  Obstetric Kit Contents:
  - Linen Saver
  - Pads
  - Sterile Gloves
  - Sterile Gauze
  - Cord Cutting Tool
  - Scalpel
  - Plastic Sheet
  - Plastic Bag

Documentation

- Identification: Record the patient’s identity, including mother and infant details if delivered, and confirm gestational age or estimated due date based on history.
- Mechanism/History: Document focused obstetric history (e.g., length of pregnancy, Gravida/Para, prenatal care, complications, drug use) and signs of labor (e.g., contraction timing, crowning, bloody show).
- Injuries/Illnesses: Note any complications (e.g., shoulder dystocia, prolapsed cord, post-partum hemorrhage) and maternal/infant conditions (e.g., hypotension, cyanosis).
- Signs/Symptoms: Record vital signs for mother (e.g., blood pressure, pulse) and APGAR scores for infant at 1 and 5 minutes post-delivery.
- Treatment/Allergies: Document interventions (e.g., controlled delivery, nuchal cord management, resuscitation, IV fluids, TXA) and any known maternal allergies or medications.
- Handover: Include time of delivery, placenta management, estimated maternal blood loss, and transfer details ensuring mother and infant go to the same facility.

Key Performance Indicators

1. Record APGAR scores at 1 and 5 minutes post-delivery.

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